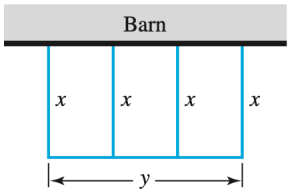
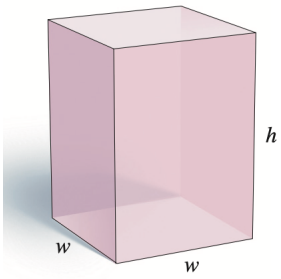
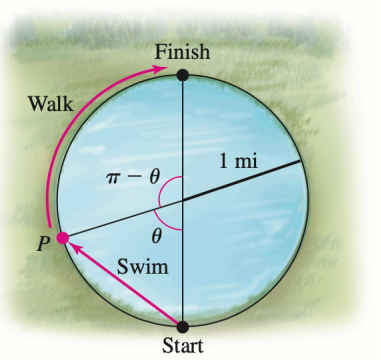
1. A rancher has 400 ft of fence for constructing a rectangular corral. One side of the corral will be formed by a barn and requires no fence. Three exterior fences and two interior fences partition the corral into three rectangular regions, as shown below. What are the dimensions of the corral that maximize the enclosed area? What is the area of the corral?



2. Suppose an airline policy states that all baggage must be box-shaped with a sum of length, width, and height not exceeding 64 in. What are the dimensions and volume of a square-based box with the greatest volume under these conditions?



3. Suppose you are standing on the short of a circular pond with a radius of 1 mile and you want to get to a point on the short directly opposite your position (on the other end of a diameter). You plan to swim at 2 mph from your current position to another point P on the shore and then walk 3 mph along the shore to the terminal point. How should you choose P to minimize the total time for the trip? (*Hint*: Start by finding the length of the chord. Note that it is the base of an isosceles triangle whose vertex angle is θ.)



4. Find the point P on the line $y = 3x$ that is closest to the point $(50,0)$. What is the least distance between P and $(50,0)$?

Answers

1. A maximum area of 10,000 ft^2 is achieved with dimensions $x = 50$ ft and $y = 200$ ft.
2. The optimal box is a cube of side length $\frac{64}{3}$ and volume 9709 in^3.
3. The *longest* time $T = 1.23$ hr is achieved when $\theta = 96°$. The shortest time is $T(\pi) = 1$ hr, when the entire trip is done swimming.
4. $(5,15)$, distance $= 47.4$ units.